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Description This document relates to Signa Horizon LX systems having a Mini-GRAM. This test is the primary means for functionally testing the Digital Tuning. This test uncovers other problems in the Gradient Driver subsystem, but focuses on the Digital Tuning hardware on the GIP Board.

1- INTRODUCTION

The Digital Tuning Test is invoked via the Diagnostics Menu found on the service page. This button is available only if the `MRconfig.cfg` file indicates digiTuningBoard = "yes". Normal calibration (GRAM Tuning, Grafidy, etc.) must be performed prior to executing this test.

Note

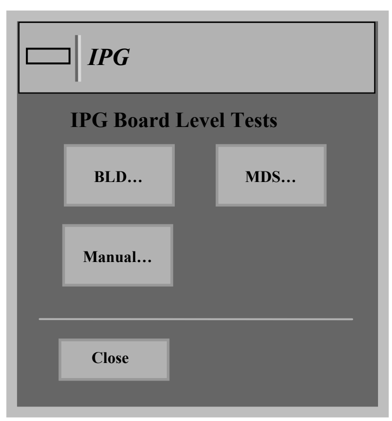
DACi ramping speed - Due to the DACi ramping speed used in the test, GRAM Overflow and Underflow overloads occur during the execution of the test. This is reflected by a flashing amber Overload LED on each GRAM Control Board. These overloads are completely normal and should be ignored. They do not cause damage to the hardware since the gradients are in Standby for this test. Software ignores these overloads and does not report them.

2- INVOKING THE DIGITAL TUNING TEST

1. Click on **[Diagnostics]** on the Service Desktop; then click **[Start]**.

2. On the Diagnostics Main Menu Screen, click on **IPG** board level box.

3. On the IPG menu (see Illustration 2-1), click on **[MDS...]**.

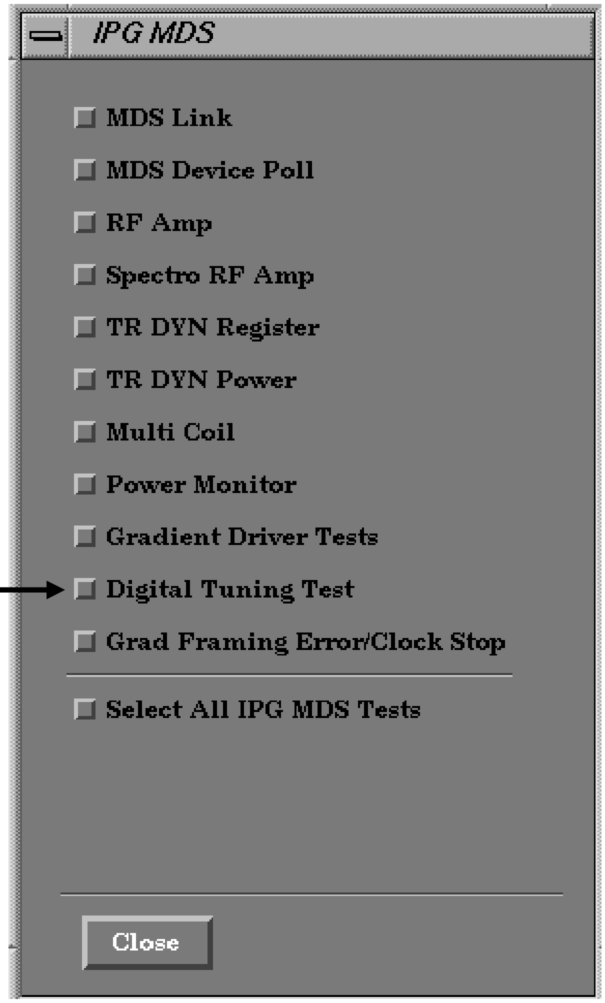


IPG SCREEN
ILLUSTRATION 2-1

4. On the IPG MDS screen (see Illustration 2-2), Select **[Digital Tuning Test]**. Then click on **[Close]** for each menu screen.

5. Click on **[Run Diags]** to start the diagnostic.

Select this test.



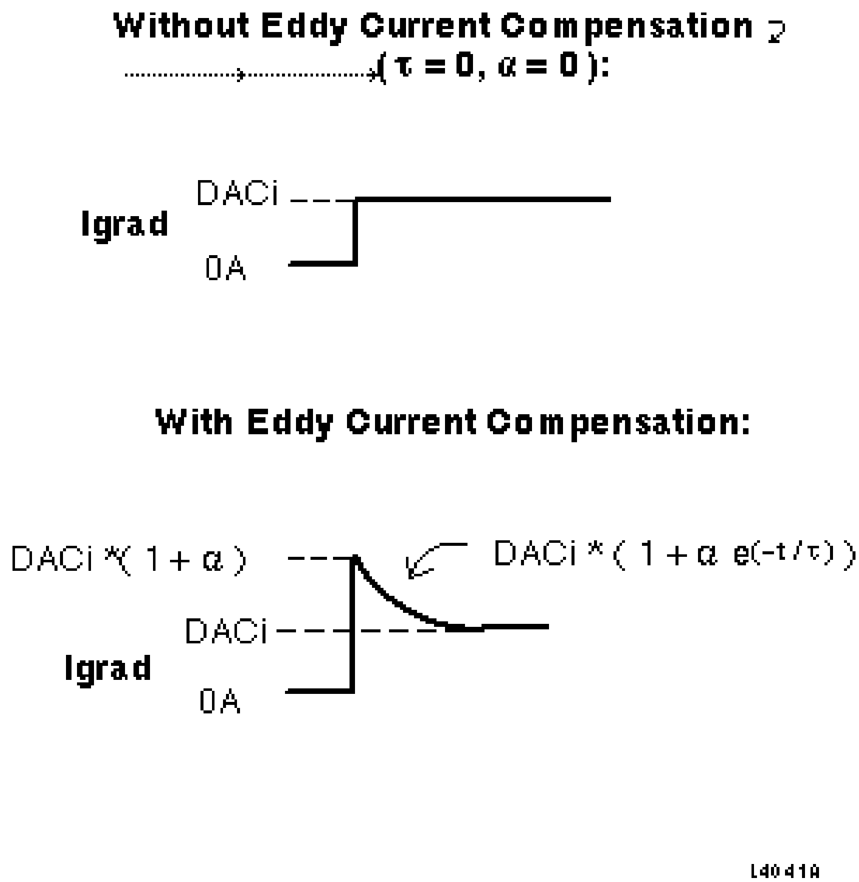
IPG MDS SCREEN
ILLUSTRATION 2-2

3- DIAGNOSTIC TEST RESULTS

Once the diagnostics have completed, data will appear indicating the faults that have occurred, if any. If a fault has occurred, remember to use the system error log to identify a recommended service strategy.

4- TEST DESCRIPTION

By cycling through a series of Digital Tuning time constants (t) and gains (a), the exponential response of the SGD System ierror signal, is examined. Illustration 4-1 shows the Exponential Response of the ierror signal with and without eddy current compensation.



EXPONENTIAL RESPONSE FOR EEC
ILLUSTRATION 4-1

The overall time constant (t) is actually composed of the summation of seven time constants. Together, these time constants provide greater resolution and overall range. Likewise, the overall gain (a) is actually composed of the summation of seven gains. Together, these gains also provide greater resolution and range.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	May 28, 1998	J. Saperstein	Initial Conversion from Toolbook to Word.
1	Feb 19, 1999	F. Fiore	Edit P. 3 to conform to bay validation.
2	Sep 30, 1999	M. Keber	Updated section 2 illustrations and procedure based on 8.3 validation.
3	Jan 19, 2000	M. Keber	Corrected section 2 test name and illustration numbers.